RELAY — WCC FORGE 48 ROUND 2 OFFICIAL SUBMISSION

Project Knowledge Reconstruction & Intelligent Handover Platform

Tagline: Never let critical project knowledge leave with people.

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1. Problem Statement

Critical project knowledge gets scattered across documents, conversations, task records, and team members. When a lead or senior engineer departs, the rationale behind decisions, unassigned dependencies, latent blockers, and unresolved questions disappear with them.

2. Solution Purpose & Identity

RELAY reconstructs usable project knowledge from authorized sources, guarantees verbatim source traceability, detects knowledge gaps across 4 primary states (Supported, Inferred, Contradicted, Missing), enforces human review, and compiles an actionable First 24 Hours Continuation Plan.

3. Technical Architecture

Frontend: React 19, Vite, Tailwind CSS v4, Lucide Icons. Backend: Node.js, Express REST API. Persistence: Dual-mode MongoDB Atlas with automatic persistent JSON storage fallback. AI Integration: OpenAI/Groq with Zod schema validation and a High-Fidelity Deterministic Demo Engine.

4. The 4 Primary Knowledge States

1. Supported: Backed by direct evidence (e.g. ADR-042 adopting TimescaleDB with 650ms benchmark).

2. Inferred: Deduced from informal discussions (e.g. HikariCP pool sizing of 120 without formal stress benchmark).

3. Contradicted: Conflicting statements across docs (e.g. Redis LRU vs LFU; Data Reconciliation 100% complete vs 32% QA test failure).

4. Missing: Actionable items lacking owners (Kafka mTLS), missing deadlines, or decisions lacking rationale (KMS 14-day key rotation).

5. The First 24 Hours Continuation Plan

Generates a time-phased operational roadmap: Hour 0-4 (Immediate triage, assign Kafka mTLS certs); Hour 4-8 (QA test failure audit, escalate FinVance API v3.4 blocker); Hour 8-16 (Reconcile Redis LRU vs LFU conflict); Hour 16-24 (PCI-DSS compliance filing briefing).

6. Testing & Quality Assurance

End-to-end user journeys fully tested and verified: document ingestion for PDF, DOCX, TXT, CSV; text extraction; gap classification; human review acceptance/rejection; First 24 Hours Continuation Plan generation; and Markdown export.

7. Three Major Round 3 Improvements

1. Continuous Knowledge Sync: Real-time integrations with GitHub, Linear/Jira, and Slack transcripts.

2. Advanced Conflict Intelligence: Deep cross-document attention and claim-level contradiction reasoning.

3. Secure Team Handover: Role-based permissions, cryptographic audit logs, and recipient-tailored views.